Lecture 18: Adjoints, Self-Adjoint, Normal

MATH 110-3

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Notation

Today V, W are finite-dimensional $\mathbb{C}$ or $\mathbb{R}$-vector spaces.

Back to Linear Maps and Operators!

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Def'n:

S'pose $T \in \mathcal{L}(V, W)$. The **adjoint** of T is the function $T^* : W \to V$ such that

$$\langle Tv, w \rangle = \langle v, T^* w \rangle$$

for every $v \in V$ and $w \in W$.

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- $T^* w$ is *that vector*

Example

Given $T : \mathbb{R}^3 \to \mathbb{R}^2$ defined $T(x_1, x_2, x_3) = (x_2 + 3x_3, 2x_1)$. Find a formula for T^*.

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Solution: $T^*(y_1, y_2) = (2y_2, y_1, 3y_1)$.

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Proof. Additivity:

$$\begin{aligned}
\langle v, T^*(w_1 + w_2)\rangle &= \langle Tv, w_1 + w2\rangle \\
&= \langle Tv, w_1\rangle + \langle Tv, w2\rangle \\
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Homogeneity also uses the proof technique of *flipping* T^* from one side of the inner product to be T on the other side.

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Prop'n [Axl14]:

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Let's prove 1,3.

Null space and range of T^*

Prop'n [Axl14]:

$T \in \mathcal{L}(V, W)$

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Proof.

$$\begin{aligned} w \in \operatorname{null} T^* &\Leftrightarrow T^* w = 0 \\ &\Leftrightarrow \langle v, T^* w \rangle = 0 \text{ for all } v \in V \\ &\Leftrightarrow \langle Tv, w \rangle = 0 \text{ for all } v \in V \\ &\Leftrightarrow w \in (\operatorname{range} T)^\perp \end{aligned}$$

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How can we conclude the rest of the proof?

Matrix of T^*

Prop'n:

$T \in \mathcal{L}(V, W)$. Orthonormal bases, $e_1, \ldots, e_n$ of V and $f_1, \ldots, f_m$ of W. Then $\mathcal{M}(T^*)$ with respect to these bases is the **conjugate transpose** of $\mathcal{M}(T)$ with these bases.

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Example:

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Example: The complex conjugate of $\begin{pmatrix} 2 & 3+4i & 7-i \\ 3 & 1 & 1-11i \end{pmatrix}$ is

$$\begin{pmatrix} 2 & 3 \\ 3-4i & 1 \\ 7+i & 1+11i \end{pmatrix}$$

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Proof.

Because the bases are orthonormal, we can read the entries of the columns of $\mathcal{M}(T)$ off of

$$Te_k = \langle Te_k, f_1 \rangle f_1 + \ldots \langle Te_k, f_m \rangle f_m$$

and $\mathcal{M}(T)_{j,k} = \langle Te_k, f_j \rangle$.

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$$T^* f_k = \langle T^* f_k, e_1 \rangle e_1 + \ldots \langle T^* f_k, e_n \rangle e_n$$

and $\mathcal{M}(T^*)_{j,k} = \langle T^* f_k, e_j \rangle = \langle f_k, Te_j \rangle = \overline{\langle Te_j, f_k \rangle} = \overline{\mathcal{M}(T)_{j,k}}$

What are the differences between T^* and T'?

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- $T^* : W \to V$, $T' : W' \to V'$
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This is definition of *Hermitian*.

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Every eigenvalue of a self-adjoint operator is real.

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Other Results

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Over $\mathbb{C}$, $\langle Tv, v \rangle$ is real for all v if and only if $T = T^*$.

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Over $\mathbb{C}$, Tv is orthogonal to v for all v if and only if $T = 0$.

Prop'n [Axl14]:

If $T = T^*$ and Tv is orthogonal to v for all v, then $T = 0$.

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$\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}$ is not self-adjoint but is normal.

Why care about normal operators?

Prop'n 7.20:

An operator T is normal if and only if $||Tv|| = ||T^*v||$ for all v.

Prop'n 7.21:

A normal operator T and its adjoint share the same eigenvectors.

Prop'n 7.22:

The eigenvectors of a normal operator of distinct eigenvalues are orthogonal.

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$$\begin{aligned} T \text{ is normal } &\Leftrightarrow T^*T - TT^* = 0 \\ &\Leftrightarrow \langle (T^*T - TT^*)v, v\rangle = 0 \text{ for all } v \in V \\ &\Leftrightarrow \langle T^*Tv, v\rangle = \langle TT^*v, v\rangle \\ &\Leftrightarrow \langle Tv, Tv\rangle = \langle T^*v, T^*v\rangle \\ &\Leftrightarrow ||Tv||^2 = ||T^*v||^2 \end{aligned}$$

In the second line, we need that $T^*T - TT^*$ is self-adjoint.

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Then

$$0 = ||(T - \lambda I)v|| = ||(T - \lambda I)^* v|| = ||(T^* - \bar{\lambda} I)v||.$$

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$$\begin{aligned}(\alpha - \beta)\langle u, v\rangle &= \langle \alpha u, v\rangle - \langle u, \bar{\beta} v\rangle \\ &= \langle Tu, v\rangle - \langle u, T^* v\rangle \\ &= 0\end{aligned}$$

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So, $\langle u, v\rangle = 0$.

References

[Axl14] Sheldon Axler.
Linear Algebra Done Right.
Undergraduate Texts in Mathematics. Springer Cham, 2014.